Introduction :

Traffic Master is a simple 3D traffic simulation project made for the Computer Graphics course. It shows a city road with cars, buses, bikes, trucks, pedestrians, buildings, trees, sidewalks and traffic lights. The project is developed using C++, OpenGL and GLUT. The main purpose is to demonstrate basic computer graphics concepts through a moving traffic scene.

Objectives :

• Create a simple 3D traffic environment.
• Use OpenGL to draw different objects.
• Show moving vehicles and pedestrians.
• Use traffic lights to control vehicles.
• Show traffic congestion.
• Use keyboard controls in the program.
• Demonstrate animation and 3D transformations.

Tools and Technologies :

Tool	Use
C++	Programming
OpenGL	3D graphics
GLUT	Window, keyboard and animation
GLU	Camera and perspective view

Main Features :

• 3D road and four-way intersection.
• Buildings, trees, grass and playground objects.
• Cars, buses, bikes and trucks.
• Moving pedestrians on sidewalks.
• Traffic lights.
• Traffic congestion percentage
• Timer and information display (HUD).
• Game Over message when traffic becomes full.

• Automatic restart after Game Over.
• Traffic sound in the background.

How the Project Works :

First, the program creates the 3D city scene. Then vehicles are added to the roads. Vehicles move in four directions and follow the traffic-light condition. Pedestrians move along the sidewalks. Page 3 The program continuously updates the positions of vehicles and pedestrians. It also checks traffic congestion and displays the current time and traffic information.

Vehicles :

The project can show up to 12 active vehicles. Different vehicle types are used: car, bus, bike and truck. Vehicles have different speeds and colors. Vehicles keep a safe distance from the vehicle in front. They stop near the intersection when their traffic direction does not have the green signal.

Pedestrians :

The project can show up to 16 pedestrians. They walk or jog on the sidewalks. Their arms and legs are rotated to create a simple walking animation.

Traffic Lights and Congestion :

The traffic light controls vehicle movement. The program also calculates traffic occupancy for the North-South and East-West roads. These values are shown on the screen. If traffic becomes too crowded, the program shows a Game Over message. The simulation can then restart automatically.

Computer Graphics Concepts Used :

• 3D modeling using basic shapes.
• Translation for moving objects.
• Rotation for vehicle and pedestrian animation.
• Scaling for changing object size.
• Perspective projection for a 3D view.
• Camera positioning using gluLookAt().
• Depth testing for correct 3D display.
• Double buffering for smooth animation.

User Controls :

Key	Action
R / r	Restart the simulation
Spacebar	Change traffic-light state
ESC	Exit the program

Expected Result :

After running the program, a 3D traffic environment appears. Vehicles move on the roads and pedestrians move on the sidewalks. Traffic lights control the vehicles. The screen shows traffic information and a timer. When traffic becomes too crowded, Game Over is displayed.

Limitations :

- The traffic behavior is simplified.
- The camera is fixed.
- Traffic lights are controlled by the user.
- The vehicle and pedestrian movements use simple rules.

Future Improvements :

- Add a movable camera.
- Add better vehicle models and textures.
- Make traffic lights automatic.
- Add collision detection.
- Add day and night effects.
- Add more roads and vehicles.

Conclusion :

Traffic Master is a simple and useful Computer Graphics project. It demonstrates 3D modeling, transformations, camera control, animation, traffic lights and user interaction. The project helped us understand how OpenGL and GLUT can be used to create a simple interactive 3D environment.